

Submission Date: **19.11.2025**

1 Classification and Regression Trees (CART): Theory

1. What are the general advantages and disadvantages of decision trees? Can you elaborate on the disadvantages and provide a short description?
2. Can you mention a few of the most important hyperparameters of the decision trees?
3. How can you make a decision tree overfit? Give three possibilities and explain.

2 Classification and Regression Trees (CART): Hands-On

1. Given the following dataset:

Ball Color	Ball Width	Sport
Yellow	6	Tennis
Yellow	6	Tennis
White	22	Football
Brown	22	Football
Brown	22	Basketball

- What is the initial split that gives the highest information gain ?
 - Build the entire decision tree based on the given dataset where, $K = 3$, $N = 5$, $V = \{\text{Tennis, Football, Basketball}\}$, $p(V) = \{\frac{2}{5}, \frac{2}{5}, \frac{1}{5}\}$
2. Assume that you are given a decision tree that splits the space as in figure 1 left, where light blue means that it assigns samples in that region to class 1 and white to class 0. Similarly, the right figure shows the ground truth, following the same color convention.
 - Determine the decision tree splits and the depth of the tree. Assume that x_1 and x_2 are the names for the horizontal and vertical axes.
 - Design a metric for assessing the quality of the prediction from the decision tree, and compute it. Consider 1 as the best-possible metric value, and 0 as worst.
 3. If we consider the ground truth region from the previous point, which would be the best split for a decision tree with depth equals to 1 (decision stump)? In other words, which is the best x that splits the region if we consider a partition like in figure 2? Try to find a formulation that is consistent with the metric specified in the last point.

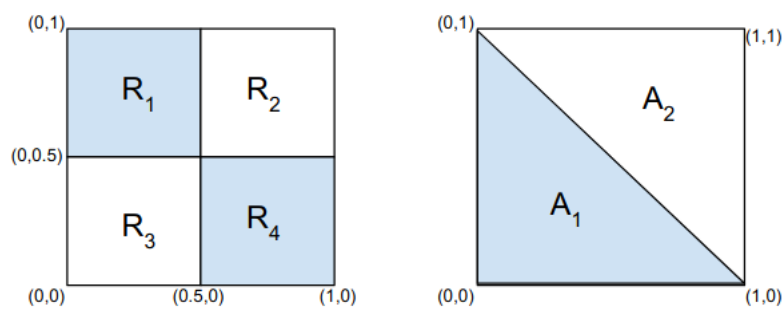


Figure 1: Figure 1. Left: Decision tree boundaries, Right: Ground truth.

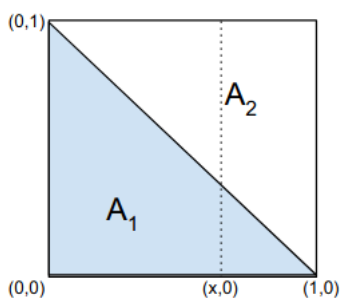


Figure 2: Figure 2. Decision Stump.

4. What would be the next split? (Hint: The next split should be on axis x_2)